

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: AFRIDOC_MT | Context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is text-only Amharic translation, matching the deployment exactly [Q1, Q44]. The benchmark provides comprehensive output-form scoring infrastructure: d-BLEU and d-chrF with bootstrap CI [Q51, Q52], chrF prioritized for African morphological richness [Q53], full results tables across models/domains/granularities [Q226–Q231], and per-prompt sensitivity analyses [Q236–Q239]. AfriCOMET is excluded from document-level scoring due to its 512-token context limit [Q175, Q176], and the long-context QE model (ModernBERT) showed no meaningful differentiation and lacks Amharic training [Q97, Q98]. GPT-4o-as-judge is explicitly documented as inconsistent for African languages including Amharic [Q99, Q218, Q219], independently corroborated [WEB-17, WEB-18]. With OF marked LOWER priority by the user (modality match), score 4: form is well-matched, but document-level neural QE for Amharic remains an open gap.

ID	Evidence
Q1	"This paper introduces AFRIDOC-MT, a document-level multi-parallel translation dataset covering English and five African languages: Amharic, Hausa, Swahili, Yorùbá, and Zulu." (p.1)
Q44	"Given that our created dataset can be used for sentence-level translation and as a baseline for document-level translation, we evaluate all models on the test splits for each domain." (p.5)
Q51	"Hence, we realigned sentence-level or pseudo-translation outputs into full documents, then computed BLEU (Papineni et al., 2002) and chrF (Popović, 2015) to create document BLEU (d-BLEU) and document chrF (d-chrF)." (p.5)
Q52	"Metrics were computed using SacreBLEU (Post, 2018) with bootstrap resampling (n = 1000) to report 95% confidence intervals." (p.5)
Q53	"We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C." (p.5)
Q97	"However, we evaluated the document-level translation outputs using ModernBERT-base-long-context-qe-v1, trained on the WMT human evaluation dataset across 41 language pairs, including over 20 languages and three African languages (Hausa, Xhosa, and Zulu), two of which are included to our work." (p.10)
Q98	"However, the scores were nearly identical across all models, offering no meaningful differentiation. Hence, for our document-level evaluation, in addition to lexical-based metrics, we incorporated three other evaluation approaches: using GPT-4o as a judge, human evaluation, and qualitative analysis." (p.10)
Q99	"GPT-4o was employed to assess and rate the translation outputs of four models. While its ratings were consistent for translations into English, the same was not observed for translations into African languages, likely due to the model's limited understanding of these languages." (p.10)
Q175	"For evaluation, we used BLEU and chrF scores but excluded AfriCOMET due to its backbone model, AfroXLM-R-L (Alabi et al., 2022; Adelani et al., 2024a), having a context length of 512 tokens." (p.22)
Q176	"This made it impractical to compute COMET scores for document-level outputs." (p.22)
Q218	"These findings are consistent with the d-chrF scores for the models, but they do not align with the evaluations from GPT-4o as a judge." (p.26)
Q219	"These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages." (p.26)
Q226	"Table 15: Performance results of various models on the sentence-level task for the Health domain, measured using document level metric d-BLEU and d-chrF." (p.29)
Q227	"Table 16: Performance results of various models on the sentence-level task for the Tech domain, measured using document level metric d-BLEU and d-chrF." (p.29)

Q228	"Table 17: Performance results of various models on the sentence-level task for the Health domain, measured using sentence level metric s-BLEU, s-CHRF, and s-COMET." (p.30)
Q229	"Performance results of various models on the sentence-level task for the Tech domain, measured using sentence level metric s-BLEU, s-CHRF, and s-COMET." (p.31)
Q230	"Table 19: Performance results of various models on the pseudo-document-level task for the Health domain, measured using document level metric d-BLEU and d-CHRF." (p.32)
Q231	"Table 20: Performance results of various models on the pseudo-document-level task for the Tech domain, measured using document level metric d-BLEU and d-CHRF." (p.32)
Q236	"Figure 19: d-chrF scores for some LLMs for sentence-level translation using different prompts when translating into African languages" (p.37)
Q237	"Figure 20: d-chrF scores for some LLMs for sentence-level translation using different prompts when translating into African languages" (p.37)
Q238	"Figure 21: d-chrF scores for some LLMs for sentence-level translation using different prompts when translating into English" (p.37)
Q239	"Figure 22: d-chrF scores for some LLMs for sentence-level translation using different prompts when translating into English" (p.37)
WEB-17	Walia-LLM corroborates GPT-4o-as-judge unreliability for Amharic (https://aclanthology.org/2024.findings-emnlp.25/)
WEB-18	arXiv 2402.08015 confirms GPT-4 'lacks ability to evaluate most generation tasks' in Amharic (https://arxiv.org/abs/2402.08015)

Strengths

- Document-level scoring (d-BLEU, d-chrF) with bootstrap-resampling 95% CIs [Q51, Q52]
- chrF prioritized for African morphological richness — sound metric choice for Amharic [Q53]
- Multiple complementary scoring streams (lexical, neural QE, GPT-4o judge, human DA, qualitative analysis) [Q98]
- Honest reporting of GPT-4o-as-judge unreliability for African languages — directs users to human DA where reliable [Q67, Q68, Q99]
- Output text-only format matches deployment's print/digital Amharic distribution requirement

ID	Evidence
Q51	"Hence, we realigned sentence-level or pseudo-translation outputs into full documents, then computed BLEU (Papineni et al., 2002) and chrF (Popović, 2015) to create document BLEU (d-BLEU) and document chrF (d-chrF)." (p.5)
Q52	"Metrics were computed using SacreBLEU (Post, 2018) with bootstrap resampling (n = 1000) to report 95% confidence intervals." (p.5)
Q53	"We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C." (p.5)
Q67	"These inconsistencies raise concerns about GPT-4o's reliability." (p.7)
Q68	"Consequently, we focus on human evaluation going forward." (p.7)
Q98	"However, the scores were nearly identical across all models, offering no meaningful differentiation. Hence, for our document-level evaluation, in addition to lexical-based metrics, we incorporated three other evaluation approaches: using GPT-4o as a judge, human evaluation, and qualitative analysis." (p.10)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text-only Amharic) matches deployment exactly. The deployment produces Amharic documents for print and digital distribution; benchmark evaluates text-only translation [Q1, Q44, infrastructure_notes].

OF-2: Text-to-speech is not in scope for this deployment — outputs are text-only. INSUFFICIENT DOCUMENTATION on TTS, but this is not a deployment requirement.

OF-3: Literacy considerations affect end-user readability. Ethiopia adult literacy ~51.8% (2017 UNESCO) [WEB-4] with significant urban-rural and gender gaps [WEB-6, WEB-7]. The benchmark's fluency dimension (1–5) and human DA (0–100) are not specifically calibrated for semi-literate populations — output text may be fluent at expert-translator level but inaccessible to lower-literacy patients. Partial mismatch in scoring calibration relative to deployment audience.

OF-4: External validity for output form is mostly preserved (text-to-text match). Two minor harms: (a) no validated document-level neural QE for Amharic [Q97, Q98, WEB-18], leaving d-chrF and human DA as the only scalable signals; (b) fluency calibration not tied to semi-literate patient comprehension [literacy_profile note].

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Q98	"However, the scores were nearly identical across all models, offering no meaningful differentiation. Hence, for our document-level evaluation, in addition to lexical-based metrics, we incorporated three other evaluation approaches: using GPT-4o as a judge, human evaluation, and qualitative analysis." (p.10)
WEB-4	Ethiopia adult literacy ~51.8% (2017 UNESCO) — relevant to readability calibration (https://www.theglobaleconomy.com/Ethiopia/Literacy_rate/)
WEB-6	Significant urban-rural and gender literacy gaps in Ethiopia (https://zoetalentsolutions.com/education-statistics-for-ethiopia/)
WEB-7	https://countrymeters.info/en/Ethiopia
WEB-18	arXiv 2402.08015 confirms GPT-4 'lacks ability to evaluate most generation tasks' in Amharic (https://arxiv.org/abs/2402.08015)

Information Gaps

- Whether deployment requires any TTS or audio output for low-literacy patient channels — unresolved in elicitation
- Whether MOH would require specific document formatting (PDF, Word, print-ready) beyond text

Requires Expert Verification

- Whether AFRIDOC-MT-style fluency scoring correlates with semi-literate-patient comprehension (could be tested with a small readability study)

Remediation

Gap 1: No validated document-level neural quality estimation metric exists for Amharic; AfriCOMET is sentence-level only and the long-context QE model excludes Amharic.

Recommendation: Adopt sentence-level AfriCOMET as a routine quality-control signal at translation-unit level, paired with d-chrF for document-level signal. Treat document-level neural QE as an open research question and avoid relying on GPT-4o-as-judge for Amharic outputs given the benchmark's own and independent corroborating evidence of unreliability [Q219, WEB-18].

ID	Evidence
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